

Voice-Controlled HOME AUTOMATION SYSTEM

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This project is about controlling home appliances with your voice through an Android app. An Arduino Mega board is used for controlling the relay through which an appliance is switched on/off. You can modify the Android app to add more functions.

First, download Bluetalk app and install it on your Android phone. Screenshot of Bluetalk app is shown in Fig. 1. Block diagram of the voice-controlled home automation system is shown in Fig. 2.

Circuit and working

Circuit diagram of the voice-controlled home automation system is shown in Fig. 3. The circuit consists of Arduino Mega 2560 board (Board1) for comparing the input string received through Bluetooth with the stored string to give output to digital pin 6 of Board1 to control the relay.



Fig. 1: Screenshot of Bluetalk app

Bluetooth module HC-05 transmits and also receives data serially via Board1 that can be read by the microcontroller (MCU). A relay is used here to control only one appliance, though you can use a multichannel relay board to control multiple appliances. Mega 2560 has 54 digital pins, each of which can be used to control an appliance. The MCU can be programmed to compare the relative strings (speech).

Software

There are two software programs used in this project. One is Arduino sketch (Bluetalk.ino) for the MCU

PARTS LIST

Semiconductors:

- Board1 - Arduino Mega 2560 board
- T1 - BC548 npn transistor
- LED1 - 5mm LED
- D1 - 1N4007 rectifier diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 1-kilo-ohm
- R2 - 680-ohm

Miscellaneous:

- RL1 - 5V, 1/CO relay
- CON1 - 6-pin connector
- CON2, CON3 - 2-pin connector terminal
- CON4 - 5-pin connector
- Bluetooth - HC-05 Bluetooth module
- Load - Electric bulb and holder
- Power - 9V battery/9V DC power supply

and the other is Android app (Bluetalk.apk).

Arduino IDE is used for programming the MCU. For developing Android app, MIT App Inventor is used, as it is easier to use as compared to Android SDK. MIT App Inventor is good for hobby projects and DIY enthusiasts to create small apps for their project-specific use.

Bluetalk.apk app uses Google's speech-recognition function and transmits the text to the MCU via the connected Bluetooth (HC-05). The .apk code can be read by importing Bluetalk.aia file in MIT App Inventor. Both .apk and .aia files are included in this month's EFY DVD.

Install Arduino IDE (v1.6.9) in your PC and connect Arduino board (here, Arduino Mega is used but we can use any other Arduino board) to the PC with the help of a serial cable. Choose correct serial COM port of the board and then upload the sketch to the MCU. Disconnect the serial cable from Arduino board and connect to the relay and Bluetooth module HC-05.

Again connect Arduino board to

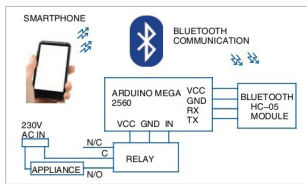


Fig. 2: Block diagram of the voice-controlled home automation system

the PC via a serial cable. An LED (not shown here) on Bluetooth module will start blinking rapidly, which means it is not connected. Once HC-05 module is connected via the smartphone, the LED will blink just twice after some delay.